

Sarvesh Pramod Datar

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Education	2024-2025	Imperial College London M.Sc. in Advanced Chemical Engineering with Process Systems Engineering Grade - Distinction Thesis : Predicting and controlling the shape of crystals for efficient API Production - A solvent selection approach Project Supervisor – Dr. Claire Adjiman
	2020-2024	MIT-WPU, Pune, India B. Tech in Chemical Engineering CGPA – 9.92 (Gold Medalist) Thesis: Process Development for production of an Active Pharmaceutical Ingredient (API) Precursor Project Supervisor – Prof. Datta Dandge
	2005 – 2020	Grade XII Score - 86.4 % Grade X Score – 94.1 %
Research experience	Sept 2025 – November 2025	University College Dublin – Intern Mentor – Dr. Sebastian Espinel Rios • Implementation of advanced process control techniques for biochemical processes • Use of reinforcement learning for PID controller tuning • Developed python programs using Pulse width modulation to control gene expression of an organism
	Jan 2024 – June 2024	IIT Bombay, India – Research Intern Advisor – Dr. Sanjay Mahajani • Catalyst screening and testing for API precursor synthesis from biomass • Catalyst characterization using ³¹P-NMR, FTIR and potentiometric titration • Process Development for sustainable production of API precursor • Process Optimization and Intensification
	June 2023 – December 2023	CSIR – National Chemical Laboratory (NCL), India – Project Trainee Advisor – Dr. Satyam Naidu Vasireddy • Kinetic Modelling of Ethylbenzene oxidation • Process Development for production of Ethylbenzene Hydroperoxide • Process Simulation of novel capillary reactors for ethylbenzene autoxidation • Study and review of catalytic oxidation processes for production of perfumery intermediate chemicals

Research Projects

Process Development for sustainably manufacturing pharmaceutical grade Formaldehyde

- Developed heuristics and process for sustainable production of formaldehyde from Carbon dioxide and green hydrogen
- Simulated a plant having annual production capacity of 70,000 tons of Formaldehyde in Aspen HYSYS
- Performed Heat integration study and optimized the production process using GAMS

Process Development for green synthesis of Copper Nanoparticles and its utilization for wastewater treatment

- Developed a process for synthesis of copper nanoparticles using Azadirachta indica leaves
- Characterized the nanoparticles using SEM, ATR-FTIR and UV-Vis Spectroscopy
- Utilized the copper nanoparticles for degradation of methylene blue dye and integrated the nanoparticles with Chitosan for development of anti-infective polymer film

Design and development of compact Medical Oxygen Concentrator

- Development of Medical Oxygen Concentrator which uses Rapid Pressure Swing Adsorption technology
- Modelling and identification of critical parameters of the adsorption process
- Optimization of adsorption process to obtain shorter cycle times, higher productivity and oxygen purity

Publications

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| 2023 | Experimental Study on Green Synthesis of Copper Nanoparticles and Its Utilization for Degradation of Organic Dye
Emerging Materials and Technologies in Water Remediation and Sensing, Springer Nature Singapore |
| 2024 | Probing the Effect of Sulfonation Density on the Acid Strength of Ion Exchange Resins using Solid-State NMR
Paper under review with Journal of Catalysis, Elsevier |

Professional training

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| 2022-2024 | Honor's Degree in Computer Science
MIT-WPU, India
Completed Honor's course and minor in Computer Science Engineering along with B. Tech in Chemical Engineering |
| 2023 | Bioreactor Design and Analysis
National Programme on Technology Enhanced Learning (NPTEL) |
| 2022 | Safety and Chemical Engineering Education (SACHE)
American Institute of Chemical Engineers (AIChE) |

Computer Skills and Softwares

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| <ul style="list-style-type: none">• HTML• Java• Python• Machine Learning• Optimisation using ML (Gradient descent, Artificial Neural Networks, Bayesian Optimisation)• Data driven Optimisation• Arduino• ThingSpeak• MIT App Inventor• Adobe Illustrator and Canva• MS Office | <ul style="list-style-type: none">• AutoCAD• Scilab• Origin• ChemDraw• MATLAB• COMSOL• ANSYS• gPROMS• GAMS• Aspen Plus and HYSYS• ChemCAD• DWSIM• GitHub |
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Languages

English, Hindi, Marathi and German